DHARANIDHARAN J

AI/ML Engineer & Backend Developer

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PROFESSIONAL SUMMARY

AI/ML engineering student and co-founder who ships production backend and AI-agent systems end-to-end. Built and deployed FastAPI services, LangGraph multi-agent pipelines, and RAG/semantic-search systems for real clients through Arcvex Studio, an agency I co-founded and where I own AI integration and backend architecture. Seeking an AI/ML or software engineering internship in Coimbatore.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL
Backend & AI Frameworks: FastAPI, LangChain, LangGraph, REST APIs, JWT Auth, PyTorch, Hugging Face
AI/ML: RAG, Multi-Agent Systems, LLM Integration, Hybrid BM25 + Semantic Search, Vector Databases
Frontend: React.js, TanStack
Databases & Infra: PostgreSQL, Supabase, MySQL, ChromaDB, SQLAlchemy ORM, Docker, Git, Linux (Pop!_OS)
Deployment & Integrations: Render, Vercel, Neon, Cloudflare, Razorpay, Shiprocket, Firebase Auth, Tauri

EXPERIENCE

Co-Founder & AI/Backend Engineer | Arcvex Studio *2026 – Present*

- Co-founded an AI-focused software agency; own FastAPI backend architecture, AI agent integration, and deployment for client web platforms, e-commerce/billing systems, and CRM tools.
- Delivered a production e-commerce platform for a client (FastAPI, React, Supabase, Razorpay payments, Shiprocket shipping automation, Firebase Auth), live on a custom domain via Vercel + Render, with a companion offline billing desktop app built in Tauri.
- Migrated a client backend from SQLite to Supabase/Postgres and hardened it: JWT + Google OAuth, Razorpay payment security fixes, and an admin panel.
- Standardized deployment into a reusable Neon + Render + Cloudflare stack, reducing setup work on each new client build.
- Led business development with two co-founders: scoping requirements, writing quotations, and closing projects, while owning technical delivery end to end.

Hackathon Competitor | Meta x PyTorch Hackathon

- Collaborated in a team to design and evaluate open-environment AI models, applying PyTorch and Hugging Face pipelines under a fixed competition timeline.

SELECTED PROJECTS

ResHub — AI-Orchestrated Developer Workspace | React, FastAPI, LangGraph, ChromaDB, Docker

- Built a multi-agent LangGraph orchestrator for managing notes, tasks, and project files through natural-language chat, with separate chat history per project. Deployed on Render + Vercel + Chroma Cloud.
- Added per-folder RAG scoping and JWT auth, with Supabase row-level security for per-user data isolation.
- Replaced keyword routing with an LLM-based JSON intent classifier, and cut the Docker image from ~8.5 GB to ~2 GB (about 75%) by removing unused CUDA dependencies.

SIA — AI-Powered Personalized Tutor | React, FastAPI, LangGraph, ChromaDB

- Built a 3-agent LangGraph pipeline that generates personalized, sequenced learning roadmaps from a user's target skills.
- Designed a three-layer dynamic context-assembly system so the tutor adapts its prompts to each learner, plus a task-tracking side panel and an in-app “Ask SIA” chatbot.
- Built a tech-news aggregator that curates articles by focus area, keeps a rolling history of the last 20 updates, and supports starring.

FileRAG — Personal AI Memory Layer | Python, BM25, Semantic Search

- Designed a personal AI memory layer using hybrid BM25 + semantic retrieval over a plain-text “soul file,” built for a local-first voice assistant. Published a write-up on DEV.to.

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning *2024 – 2028*
Sri Shakthi Institute of Engineering and Technology, Coimbatore | CGPA: 7.8 / 10.0